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ON THE DIRECTIONS OCCURRING IN LATTICE-LINE COVERINGS OF THE INTEGER PLANE

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Abstract. We consider families of lines that cover every point of the integer lattice $\mathbb{Z}^2$, subject to the constraint that no two lines of different direction in the family meet at a lattice point. Restricting to *lattice lines* (lines containing at least two, hence infinitely many, lattice points, equivalently of rational direction), we show that the set of directions occurring in such a covering can be made dense in the space of line directions. The construction is a recursive splitting of $\mathbb{Z}^2$ into nested rank-2 sublattice cosets, each handed off to a freshly chosen direction; the key technical point is a steering lemma showing that at every stage of the recursion a new direction arbitrarily close to any prescribed target can still be realized, via an elementary sieve bound.

1. Introduction

Consider the integer lattice $\mathbb{Z}^2 \subset \mathbb{R}^2$. We wish to cover every point of $\mathbb{Z}^2$ by a family $\mathcal{F}$ of lines, subject to: any two lines of $\mathcal{F}$ may cross, but never at a point of $\mathbb{Z}^2$. Which sets of line-directions can occur across such a family?

Remark 1 (The lattice-line hypothesis). As posed, with no further restriction on the lines allowed, the question is trivial: for each $z \in \mathbb{Z}^2$ take the line L_z through z of irrational slope α_z. Then

$$L_z \cap \mathbb{Z}^2 = \{z\} \tag{1}$$

(a second lattice point on L_z would force $\alpha_z \in \mathbb{Q}$), so distinct such lines never share a lattice point, and choosing the α_z to range over a dense (indeed, all) set of irrational directions already realizes an uncountable dense set of directions. Everything of interest in this note therefore concerns the restriction to *lattice lines*: lines containing at least two (hence, by the same argument, infinitely many) points of $\mathbb{Z}^2$, equivalently lines of rational direction. This restriction is in force throughout.

We now formalize the objects of study as explicit definitions.

Definition 2 (Primitive direction)**.** A *primitive direction* is a pair $d = (p, q) \in \mathbb{Z}^2$ with $\gcd(p, q) = 1$. Two primitive directions d, d' are *equal as directions* if $d' = \pm d$. Write

$$\varphi_d(x, y) = qx - py, \tag{2}$$

a surjective homomorphism $\mathbb{Z}^2 \to \mathbb{Z}$ depending only on d up to sign of the whole map (in particular $\ker \varphi_d = \ker \varphi_{-d}$). The space of directions is

$\mathbb{RP}^1 = S^1/\{\pm 1\}$, parametrized by $\theta \in [0, \pi)$; we metrize it by

$$d(\theta, \theta') = \min(|\theta - \theta'|,\ \pi - |\theta - \theta'|), \tag{3}$$

so "within ε of θ" always means within ε in this metric. Since $[0, \pi) \to \mathbb{RP}^1$ is a continuous surjection, a set dense in $[0, \pi)$ is automatically dense in $\mathbb{RP}^1$.

Definition 3 (Lattice lines)**.** For a primitive direction $d = (p, q)$ and $c \in \mathbb{Z}$, the *lattice line* $\ell_{d,c}$ is the level set

$$\ell_{d,c} \;=\; \varphi_d^{-1}(c) \;=\; \{(x, y) \in \mathbb{Z}^2 : qx - py = c\}. \tag{4}$$

Every line in $\mathbb{R}^2$ containing at least two points of $\mathbb{Z}^2$ equals $\ell_{d,c}$ for a unique primitive direction d (up to sign) and a unique $c \in \mathbb{Z}$; we call such a line a lattice line of direction d.

Definition 4 (Covering family, valid family)**.** A *covering family* is a set $\mathcal{F}$ of lattice lines with $\bigcup \mathcal{F} \supseteq \mathbb{Z}^2$. A covering family $\mathcal{F}$ is *valid* if no two lines of $\mathcal{F}$ of different direction share a point of $\mathbb{Z}^2$; equivalently, for every pair of distinct primitive directions d, d' occurring in $\mathcal{F}$, and every $\ell_{d,c}, \ell_{d',c'} \in \mathcal{F}$, we have $\ell_{d,c} \cap \ell_{d',c'} \cap \mathbb{Z}^2 = \emptyset$.

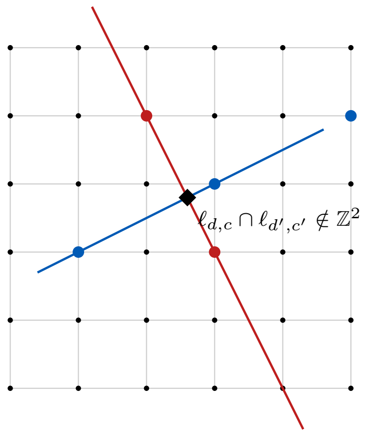

FIGURE 1. Two lattice lines of different directions d, d' may cross *off* the lattice (diamond marker) without violating validity (Definition 4): only crossings *at* points of $\mathbb{Z}^2$ (dots) are forbidden between distinct directions.

Lemma 5 (The everywhere-disjoint variant is trivial)**.** *If the lines of $\mathcal{F}$ are required to be pairwise disjoint everywhere (not merely off the lattice), then every line in $\mathcal{F}$ shares a single direction, and any one direction already suffices (Example 6 below).*

Proof. Two distinct lines in $\mathbb{R}^2$ are either parallel or meet at exactly one point, so pairwise disjointness forces every line in the family to share a single direction. □

All the content is in the "crossings allowed off the lattice" variant of Definition 4, which is our subject here.

Example 6. For a primitive direction $d = (p, q)$, the lines $\ell_{d,c}$, $c \in \mathbb{Z}$, partition $\mathbb{Z}^2$ (they are the fibers of the surjective homomorphism φ_d):

$$\mathbb{Z}^2 \;=\; \bigsqcup_{c\in\mathbb{Z}} \ell_{d,c}. \tag{5}$$

So a single direction always suffices to cover $\mathbb{Z}^2$ by pairwise-disjoint (in particular lattice-point-disjoint) lattice lines.

Our main result:

Theorem 7. *There is a valid covering family $\mathcal{F}$ (Definition 4) whose set of realized directions is dense in $\mathbb{RP}^1$.*

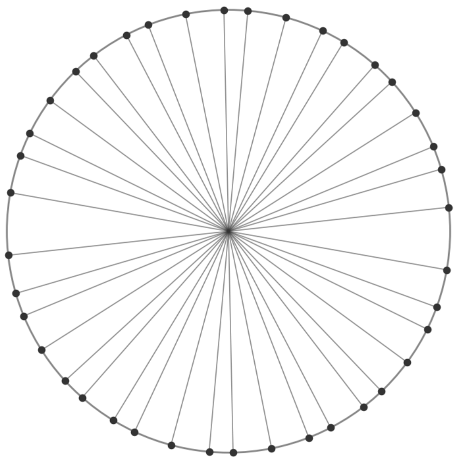

FIGURE 2. A preview of Theorem 7: the first 100 directions realized by the construction of Section 3 below, each drawn as a diameter of the unit circle. See Figure 3 for the same data with the construction's progress (order and consecutive jumps) made visible.

2. Two lemmas on lattice lines

Lemma 8 (Rigidity). *Let $d_1 = (p_1, q_1)$, $d_2 = (p_2, q_2)$ be* two *distinct primitive directions, and set*

$$\Delta \;=\; p_1q_2 - p_2q_1 \;\neq\; 0. \tag{6}$$

The lines ℓ_{d_1,c_1} and ℓ_{d_2,c_2} meet at a lattice point if and only if

$$\Delta \mid (p_1c_2 - p_2c_1) \qquad \textit{and} \qquad \Delta \mid (q_1c_2 - q_2c_1). \tag{7}$$

In particular, if $|\Delta| = 1$, condition (7) *holds vacuously for* every *choice of c_1, c_2, so d_1, d_2 can never both occur in a valid family (Definition 4).*

Proof. A lattice point (x, y) lies on both lines exactly when the linear system

$$\begin{aligned} q_1x - p_1y &= c_1,\\ q_2x - p_2y &= c_2 \end{aligned} \tag{8}$$

holds. Eliminating y gives

$$\Delta\, x \;=\; p_1c_2 - p_2c_1, \tag{9}$$

and eliminating x symmetrically gives

$$\Delta\, y \;=\; q_1c_2 - q_2c_1. \tag{10}$$

Since $\Delta \neq 0$, equations (9)–(10) determine $x, y \in \mathbb{Q}$ uniquely (this is exactly Cramer's rule applied to (8)), and $(x, y) \in \mathbb{Z}^2$ if and only if Δ divides both right-hand sides, which is precisely (7). Conversely, if (7) holds, the quotients in (9)–(10) are integers and one checks directly, by substituting back into (8), that they solve the system. If $|\Delta| = 1$, both divisibility conditions in (7) hold automatically for every $c_1, c_2 \in \mathbb{Z}$. □

Lemma 9 (Coset)**.** *Let p, q be coprime integers, $x_0, y_0 \in \mathbb{Z}$, and set*

$$c_0 = qx_0 - py_0. \tag{11}$$

For $k \in \mathbb{Z}$, write $r_k = c_0 + pqk$. Then

$$\bigcup_{k\in\mathbb{Z}} \ell_{(p,q),\, r_k} \;=\; \{(x, y) \in \mathbb{Z}^2 : x \equiv x_0 \pmod{|p|},\ y \equiv y_0 \pmod{|q|}\}. \tag{12}$$

Proof. Write C for the right-hand side of (12). Since $\gcd(p, q) = 1$ implies in particular $p, q \neq 0$, congruence modulo p (resp. q) is well defined, and C admits the parametrization

$$C = \{(x_0 + pu,\ y_0 + qv) : u, v \in \mathbb{Z}\}. \tag{13}$$

For a point of the form (13),

$$\varphi_{(p,q)}(x_0 + pu,\ y_0 + qv) \;=\; q(x_0 + pu) - p(y_0 + qv) \;=\; c_0 + pq(u - v), \tag{14}$$

using (11) and cancelling the pqu terms. By (4), this point lies on $\ell_{(p,q),r_k}$ if and only if $u - v = k$. As u ranges over $\mathbb{Z}$ (with $v = u - k$ then determined), every point of C with that particular value of $u - v$ is visited exactly once, and (14) shows every point of $\ell_{(p,q),r_k} \cap C$ has $u - v = k$: hence

$$\ell_{(p,q),\, r_k} \cap C \;=\; \{(x_0 + pu,\ y_0 + q(u - k)) : u \in \mathbb{Z}\}, \tag{15}$$

and moreover every point of $\ell_{(p,q),r_k}$ manifestly lies in C (take u, v with $u - v = k$ so that (14) matches r_k; such (x, y) automatically satisfies $x \equiv x_0 \pmod{p}$, $y \equiv y_0 \pmod{q}$). So $\ell_{(p,q),r_k} \subseteq C$ for every k, with equality of the union to all of C once k ranges over $\mathbb{Z}$, since every pair $(u, v) \in \mathbb{Z}^2$ has *some* value of $u - v$. □

Recall that by *Farey neighbors* is meant two primitive directions

$$d_1 = (p_1, q_1), \quad d_2 = (p_2, q_2)$$

(equivalently, fractions p_1/q_1, p_2/q_2 in lowest terms) with

$$|p_1q_2 - p_2q_1| = 1.$$

This is exactly the $|\Delta| = 1$ case of Lemma 8; see [4, Ch. III] for the classical theory. The rigidity lemma shows that a family realizing many directions cannot simply throw them together: directions related by a determinant of ± 1 ("unimodular pairs", e.g. any pair of Farey neighbors read as slopes) are permanently incompatible. The coset lemma is the tool that lets us build a family avoiding all bad pairs at once: applied recursively below, it splits the current sub-coset into finer sub-cosets, one of which is handed off to each freshly chosen direction.

3. The construction

Definition 10 (Rank-2 coset)**.** For coprime positive integers n_1, n_2 and $x_0, y_0 \in \mathbb{Z}$, define

$$R(n_1, n_2, x_0, y_0) \;=\; \{(x, y) \in \mathbb{Z}^2 : x \equiv x_0 \pmod{n_1},\; y \equiv y_0 \pmod{n_2}\}. \tag{16}$$

Since $n_1, n_2 > 0$, every $(x, y) \in R(n_1, n_2, x_0, y_0)$ is of the form $x = x_0 + n_1 u$, $y = y_0 + n_2 w$ for a *unique* pair $(u, w) \in \mathbb{Z}^2$; call (u, w) the *internal coordinates* of the point.

We state the two halves of the original Splitting lemma as two separate lemmas: the first identifies each finer sub-coset with a plain coset of a new direction, and the second records that these sub-cosets genuinely partition R.

Lemma 11 (Splitting: coset identity)**.** *Let $R = R(n_1, n_2, x_0, y_0)$, let s, t be nonzero integers with*

$$\gcd(s, n_2) = \gcd(t, n_1) = \gcd(s, t) = 1, \tag{17}$$

and put

$$P = n_1 s, \qquad Q = n_2 t. \tag{18}$$

For $u_0, w_0 \in \mathbb{Z}$, let

$$\begin{aligned} R_{u_0,w_0} \;=\; \{(x, y) \in R : x = x_0 + n_1 u,\; y = y_0 + n_2 w \text{ for some} \\ u \equiv u_0 \pmod{|s|},\; w \equiv w_0 \pmod{|t|}\} \end{aligned} \tag{19}$$

(a sub-coset of R, picked out via the internal coordinates of Definition 10). Then

$$\bigcup_{k\in\mathbb{Z}} \ell_{(P,Q),\, c_0+PQk} \;=\; R_{u_0,w_0}, \qquad c_0 = Q(x_0 + n_1 u_0) - P(y_0 + n_2 w_0). \tag{20}$$

In particular (P, Q) is itself a primitive direction: $\gcd(P, Q) = 1$.

Proof. Primitivity of (P, Q). A prime dividing both $P = n_1 s$ and $Q = n_2 t$ would have to divide one of n_1, n_2 (excluded, as $\gcd(n_1, n_2) = 1$), or n_1 and t (excluded by $\gcd(t, n_1) = 1$ in (17)), or s and n_2 (excluded by $\gcd(s, n_2) = 1$), or s and t (excluded by $\gcd(s, t) = 1$). These four cases are exhaustive, so no such prime exists.

The identity (20). For $(x, y) = (x_0 + n_1 u,\, y_0 + n_2 w)$,

$$Qx - Py \;=\; Q(x_0 + n_1 u) - P(y_0 + n_2 w) \;=\; (Qx_0 - Py_0) + n_1 n_2 (tu - sw), \tag{21}$$

using $Q = n_2 t$, $P = n_1 s$. Put $c_0^{\mathrm{loc}} = tu_0 - sw_0$. Then, by Lemma 9 applied to the coprime pair (s, t) in the *internal* coordinates (u, w),

$$\bigcup_{k\in\mathbb{Z}} \{(u, w) : tu - sw = c_0^{\mathrm{loc}} + stk\} \;=\; \{u \equiv u_0 \pmod{|s|},\, w \equiv w_0 \pmod{|t|}\}. \tag{22}$$

Substituting the local level $c_0^{\mathrm{loc}} + stk$ into (21) via x_0, y_0 translates it into the global level

$$\begin{aligned}(Qx_0 - Py_0) + n_1 n_2\bigl(c_0^{\mathrm{loc}} + stk\bigr) &= c_0 + n_1 n_2 st\,k \\ &= c_0 + PQk,\end{aligned} \tag{23}$$

where the first equality uses $c_0 = (Qx_0 - Py_0) + n_1 n_2 c_0^{\mathrm{loc}}$ (a direct computation confirming this matches the formula for c_0 stated in (20)), and the second uses $PQ = n_1 n_2 st$. Combining (22) and (23) with (21) gives exactly (20). □

Example 12. Take $n_1 = 2$, $n_2 = 3$, $x_0 = y_0 = 1$, so

$$R = R(2,3,1,1) = \{(x,y) : x \text{ odd},\ y \equiv 1 \pmod 3\}.$$

Take $s = 2$, $t = 3$; one checks directly that (17) holds:

$$\gcd(s, n_2) = \gcd(2,3) = 1, \tag{24}$$

$$\gcd(t, n_1) = \gcd(3,2) = 1, \tag{25}$$

$$\gcd(s, t) = \gcd(2,3) = 1. \tag{26}$$

Then $P = n_1 s = 4$, $Q = n_2 t = 9$ (and indeed $\gcd(4,9) = 1$). Take the residue pair $(u_0, w_0) = (1,2)$; then

$$c_0 = Q(x_0 + n_1 u_0) - P(y_0 + n_2 w_0) = 9\cdot(1+2) - 4\cdot(1+6) = 27 - 28 = -1. \tag{27}$$

Three points of $R_{1,2}$, at three different values of k in (20):

$$\begin{aligned}&(u,w) = (1,2) \longmapsto (x,y) = (3,7) : \\ &\qquad \varphi_{(4,9)}(3,7) = 27 - 28 = -1 = c_0 + 36 \cdot 0, \qquad (28)\\ &(u,w) = (3,2) \longmapsto (x,y) = (7,7) : \\ &\qquad \varphi_{(4,9)}(7,7) = 63 - 28 = 35 = c_0 + 36 \cdot 1, \qquad (29)\\ &(u,w) = (1,5) \longmapsto (x,y) = (3,16) : \\ &\qquad \varphi_{(4,9)}(3,16) = 27 - 64 = -37 = c_0 + 36 \cdot (-1), \qquad (30)\end{aligned}$$

using $PQ = 36$ throughout.

Lemma 13 (Splitting: partition)**.** *With R, s, t as in Lemma 11,*

$$R = \bigsqcup_{\substack{0 \le u_0 < |s| \\ 0 \le w_0 < |t|}} R_{u_0, w_0} \tag{31}$$

(a disjoint union over the $|s||t|$ residue pairs).

Proof. Fix $(x,y) \in R$, with internal coordinates (u,w) as in Definition 10. By the division algorithm,

$$u = s\left\lfloor \frac{u}{s} \right\rfloor + u_0, \qquad 0 \le u_0 < |s|,$$

for a unique such u_0 (u_0 is the unique representative of u modulo s lying in $[0, |s|)$, and similarly $w_0 \in [0, |t|)$ for w modulo t). By construction,

$$u \equiv u_0 \pmod{|s|} \quad \text{and} \quad w \equiv w_0 \pmod{|t|},$$

so $(x, y) \in R_{u_0,w_0}$ for this particular pair (u_0, w_0), establishing

$$R \subseteq \bigcup_{u_0,w_0} R_{u_0,w_0};$$

the reverse inclusion is immediate from (19). For disjointness (equivalently, uniqueness of (u_0, w_0) for each point): suppose

$$(x, y) \in R_{u_0,w_0} \cap R_{u'_0,w'_0}$$

via internal coordinates (u, w) and (u'', w'') respectively. Since internal coordinates are unique (Definition 10), $u'' = u$ and $w'' = w$, so

$$u_0 \equiv u \equiv u'_0 \pmod{|s|};$$

as both $u_0, u'_0 \in [0, |s|)$, an interval of length $|s|$, and their difference is a multiple of s of absolute value $< |s|$, we get $u_0 = u'_0$, and symmetrically $w_0 = w'_0$. □

A naive approach to this construction might attempt to realize an *arbitrary* target direction by using a multiple of it known to lie in the current sublattice, without checking that the multiple's own geometric line (which follows its *primitive* direction, not the multiple used to construct it) stays confined to the intended coset; this is exactly the pitfall Lemma 11 avoids, via the primitivity argument built into its own proof.

Lemma 14 (Steering)**.** *Let n_1, n_2 be coprime positive integers. For every $\theta \in [0, \pi)$ and every $\varepsilon > 0$ there exist nonzero integers s, t, $|s|, |t| \geq 2$, satisfying* (17)*, such that the direction of $(n_1 s, n_2 t)$ lies within ε of θ.*

Proof. By symmetry (swap the roles of s and t to treat $\theta = \pi/2$, which targets reciprocal slope 0 and is handled by the same argument below with the roles of n_1, n_2 exchanged, subject to the same window shift introduced at (36)) it suffices to treat $\theta \neq \pi/2$, i.e. a finite target slope $\mu = \tan\theta$; the case $\mu < 0$ is identical with $t \mapsto -t$, so assume $\mu \geq 0$.

Set

$$\delta \;=\; \frac{1}{2}\prod_{r|n_1}\Big(1 - \frac{1}{r}\Big) \;>\; 0, \qquad E \;=\; 2^{\omega(n_1)+1}, \tag{32}$$

both depending only on n_1, where $\omega(n_1)$ is the number of distinct prime divisors of n_1. For a prime $p \nmid n_1 n_2$ and any interval $I \subset \mathbb{Z}$ of length L, inclusion–exclusion over the divisors of $n_1 p$ gives

$$\#\{t \in I : \gcd(t, n_1 p) = 1\} \;\geq\; L\prod_{r|n_1}\Big(1 - \frac{1}{r}\Big)\Big(1 - \frac{1}{p}\Big) - 2^{\omega(n_1)+1} \;\geq\; \delta L - E, \tag{33}$$

using $p \geq 2$ for the last factor. Fix

$$L_0 = \left\lceil \frac{E+1}{\delta} \right\rceil, \tag{34}$$

independent of p — and, crucially, independent of μ as well. Choose a prime $p \nmid n_1 n_2$ large enough that

$$L_0 + 2 \;<\; \frac{\varepsilon n_1 p}{n_2} \tag{35}$$

(possible since the right side $\to \infty$ as $p \to \infty$, for fixed ε, n_1, n_2; note (35) does not involve μ at all, unlike t_0 below), set $s = p$ and

$$t_0 = \left\lfloor \frac{\mu n_1 p}{n_2} \right\rfloor + 2. \tag{36}$$

The $+2$ shift in (36), absent from an earlier version of this proof, is what makes the forthcoming $|t| \geq 2$ requirement hold *unconditionally* rather than only "for p large": since $\mu \geq 0$, already $t_0 \geq 2$ by (36), so every integer in the window I below inherits $|t| \geq 2$ directly, with no degenerate behaviour at $\mu = 0$ (where, without the shift, $t_0 = 0$ for *every* p, the window never moves away from the origin as $p \to \infty$, and $t = 1$, always coprime to $n_1 p$, is a witness that nothing rules out being the *only* one, violating $|t| \geq 2$). The interval $I = [t_0, t_0 + L_0)$ then contains, by (33) with $L = L_0$ and (34), at least $\delta L_0 - E \geq 1$ integer t with $\gcd(t, n_1 p) = 1$ — equivalently $\gcd(t, n_1) = \gcd(t, s) = 1$, as $s = p$ is prime — and, as just noted, $|t| \geq 2$ automatically. For any such t,

$$\left| \frac{n_2 t}{n_1 s} - \mu \right| \; = \; \frac{n_2}{n_1 p} \left| t - \frac{\mu n_1 p}{n_2} \right| \; < \; \frac{n_2 (L_0 + 2)}{n_1 p} \; < \; \varepsilon, \tag{37}$$

using (35) for the last step and $\left| t - \mu n_1 p / n_2 \right| < L_0 + 2$, which follows from $t \in [t_0, t_0 + L_0)$ and $t_0 - \mu n_1 p / n_2 \in (1, 2]$ by (36), so $(P, Q) = (n_1 s, n_2 t)$ has slope within ε of μ, hence (as arctan is 1-Lipschitz, and $\arctan \mu = \theta - \pi$ when $\theta > \pi/2$ identifies the same point of $\mathbb{RP}^1$) direction within ε of θ. $\square$

The point of Lemma 14 is that it is *not* enough to know that *some* admissible (s, t) exists at every stage: repeatedly reusing, say, $(s, t) = (2, 3)$ is admissible forever (all three coprimality conditions of (17) hold trivially against the resulting $n_1 = 2^k, n_2 = 3^k$), yet gives directions converging to a single point of $\mathbb{RP}^1$, not a dense set. What is needed, and what Lemma 14 provides, is the ability to steer toward an *arbitrary prescribed* target at every stage.

Remark 15 (Tails of a dense sequence remain dense). If X is a T_1 topological space with no isolated points, $D \subseteq X$ is dense, and $F \subset X$ is finite, then $D \setminus F$ is still dense in X. This follows since any nonempty open U is itself infinite, as no point of X is isolated; so $U \setminus F$ is nonempty and open, hence meets D, giving a point of $D \setminus F$ in U. Applied with $X = \mathbb{RP}^1$, which is compact, metric, hence T_1, and has no isolated points (it is a circle), and $D = \{\theta_k\}_{k \geq 1}$: every tail $\{\theta_k\}_{k \geq N}$ is dense, since it contains $D \setminus \{\theta_1, \ldots, \theta_{N-1}\}$. This is the fact the density argument below needs beyond mere density of $\{\theta_k\}$ itself — without "no isolated points" it can fail (e.g. a sequence dense in $\{0\} \cup [1, 2]$ that visits 0 only once).

We now have all pieces needed to prove Theorem 7.

Proof of Theorem 7. Fix an enumeration of $\mathbb{Z}^2$, $(z_k)_{k \geq 1}$, and a sequence $(\theta_k)_{k \geq 1}$ dense in $[0, \pi)$ (e.g. $\theta_k = k\alpha \bmod \pi$ for α/π irrational, by equidistribution of the irrational rotation). Set $R_0 = \mathbb{Z}^2$, i.e. $(n_1, n_2, x_0, y_0) = (1, 1, 0, 0)$.

Given $R_{k-1} = R(n_1, n_2, x_0, y_0)$, apply Lemma 14 to find s, t (possibly negative — Lemma 14 allows this, needed to reach both signs of slope) realizing a direction within $1/k$ of θ_k, and apply Lemmas 11–13 to split R_{k-1} into its $|s||t| \geq 4$ sub-cosets, each fully coverable by direction $(P_k, Q_k) = (n_1 s, n_2 t)$ restricted to one residue class of levels. Put into $\mathcal{F}$ all such lines for every sub-coset except one, reserved as follows: if $z_k \in R_{k-1}$, reserve the lexicographically-least (u_0, w_0) whose sub-coset does not contain z_k (possible since there are ≥ 4 sub-cosets and only one need be avoided); otherwise reserve the lexicographically-least (u_0, w_0) outright. A concrete rule is needed for the recursion to be a well-defined function, as is a concrete choice among the (s, t) that Lemma 14 only asserts to exist. Both are choice functions to be supplied explicitly in a Lean formalization, not just existence claims. Let $R_k = R_{u_0,w_0}$ be that reserved sub-coset.

The two quantities that matter going forward are genuinely different: the *achieved direction* recorded for the density argument below is the (possibly signed) pair $(P_k, Q_k) = (n_1 s, n_2 t)$, while R_k itself, to serve as the input $R(n_1', n_2', x_0', y_0')$ to the *next* stage's application of Lemma 14 (whose hypotheses require positive n_1', n_2', as does Definition 10), must use the *positive* pair

$$n_1' = n_1|s|, \quad n_2' = n_2|t|, \quad x_0' = x_0 + n_1 u_0, \quad y_0' = y_0 + n_2 w_0 :$$

a direct check from the definition of R_{u_0,w_0} (19) (writing $u = u_0 + |s|u'$, $w = w_0 + |t|w'$ for the internal coordinates, since "$u \equiv u_0 \pmod{|s|}$" means exactly this) shows $R_{u_0,w_0} = R(n_1|s|, n_2|t|, x_0', y_0')$ on the nose.

Write $R_k = R(n_1^{(k)}, n_2^{(k)}, x_0^{(k)}, y_0^{(k)})$ for this representation, i.e. $n_1^{(k)} = n_1|s|$, $n_2^{(k)} = n_2|t|$ (with $n_1^{(0)} = n_2^{(0)} = 1$ from the base case), the notation used below. This keeps

$$\gcd(n_1^{(k)}, n_2^{(k)}) = \gcd(n_1|s|, n_2|t|) = \gcd(P_k, Q_k) = 1$$

(Lemma 11), so the invariant needed for the next iteration is preserved.

Coverage invariant. Stage i claims exactly $R_{i-1} \setminus R_i$, and $R_i \subseteq R_{i-1}$ for every $i \geq 1$. Telescoping from $R_0 = \mathbb{Z}^2$ gives, for every k,

$$\mathbb{Z}^2 \setminus R_{k-1} \;=\; \bigcup_{i<k} (R_{i-1} \setminus R_i) \;=\; \bigcup_{i<k} (\text{stage-}i\text{ claims}), \tag{38}$$

i.e. a point lies outside the current region R_{k-1} exactly when some earlier stage already claimed it. This is what justifies the reservation rule above only ever needing to avoid z_k when $z_k \in R_{k-1}$: whenever $z_k \notin R_{k-1}$, (38) shows z_k was already claimed by an earlier stage, so no action is needed for it at stage k.

Coverage. By (38), every z_k is eventually claimed: if $z_k \in R_{k-1}$, the reservation rule ensures z_k lies in one of the claimed sub-cosets at stage k (not the reserved R_k); if $z_k \notin R_{k-1}$, it was already claimed at some earlier stage. So

$$\bigcup_k (\text{stage-}k\text{ claims}) \;=\; \mathbb{Z}^2. \tag{39}$$

Disjointness. Every individual line placed into $\mathcal{F}$ at stage k has all of its lattice points inside the sub-coset R_{u_0,w_0} it came from (a subset of $R_{k-1} \setminus R_k$, since $R_{u_0,w_0} \neq R_k$): this is exactly (20) of Lemma 11 (each line on the left is a subset of the union), the step the earlier, flawed version of this construction got wrong (see the remark following Lemma 11). So stage k's claimed points lie in

$$R_{k-1} \setminus R_k \subset R_{k-1}.$$

For $j > k$, $R_{j-1} \subseteq R_k$ (as $R_i \subseteq R_{i-1}$ for every i), so stage j's points, lying in R_{j-1}, are disjoint from stage k's. Hence distinct lines from different stages never share a lattice point, and distinct lines of the same direction from the same stage are disjoint level sets of the same $\varphi_{(P_k,Q_k)}$, so never share one either. (Different stages in fact always realize different directions: $|P_k| = n_1^{(k)}$ is strictly increasing in k, since

$$n_1^{(k)} = n_1^{(k-1)}|s_k| \geq 2n_1^{(k-1)};$$

though the argument above does not need this.)

Density. Fix $\theta \in \mathbb{RP}^1$ and $\varepsilon > 0$. By Remark 15, every tail $\{\theta_k\}_{k \geq N}$ is dense in $\mathbb{RP}^1$; so for any N there is some $k \geq N$ with θ_k within $\varepsilon/2$ of θ. Taking $N > 2/\varepsilon$, the corresponding achieved direction (P_k, Q_k) — within $1/k < \varepsilon/2$ of θ_k — then lies within ε of θ. (Each (P_k, Q_k) is genuinely realized in $\mathcal{F}$: at least $|s||t| - 1 \geq 3$ of the ≥ 4 sub-cosets at stage k are claimed, all nonempty, each contributing lines of that direction.) So every $\theta \in \mathbb{RP}^1$ is a limit of directions realized in $\mathcal{F}$: the realized direction set is dense.

$\square$

4. A picture of the construction

Figure 3 plots the first 10, 20, and 100 directions realized by an actual run of the construction (code in `code/`, targeting the golden-angle equidistributed sequence $\theta_k = k\pi(\sqrt{5} - 1) \bmod \pi$), each direction shown as an antipodal pair of points on the unit circle (a line's direction is unsigned) and colored by its position in the sequence.

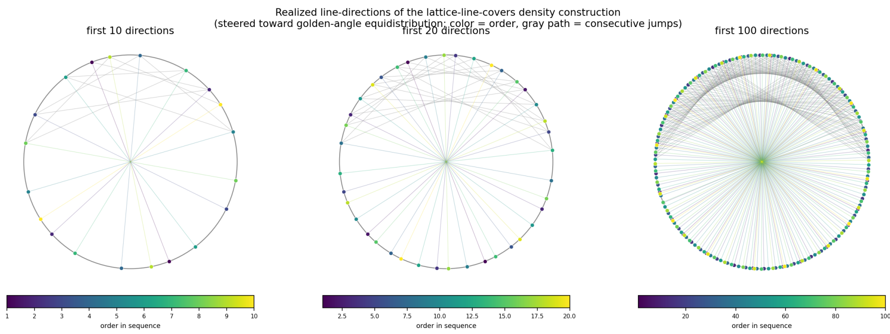

Figure 3. Realized directions after 10, 20, and 100 steps of the construction, colored by order (dark = early, light = late), with a thin path tracing consecutive jumps. The steering argument scatters the directions rather than sweeps them around the circle, since each target angle is chosen from an equidistributed sequence rather than visited in angular order.

Remark 16. Both the splitting mechanism (Lemmas 11–13) and the sign-generality of Lemma 14 were checked by direct brute-force simulation (exhaustive crossing and coverage checks on finite windows up to 81×81, including adversarial diagonal and sign-alternating schedules) before being trusted; an earlier version of the splitting step, without the primitivity safeguard built into Lemma 11, was found this way to be genuinely flawed rather than merely under-justified. The full argument was also read independently by a second reader (another Claude agent) working from the statement of the lemmas alone, who found a real gap in an earlier version of the density argument (restricting s, t to be positive only reaches half of $\mathbb{RP}^1$, and existence of *some* admissible move at each stage does not imply the ability to steer toward an *arbitrary* target — both addressed in the versions of Lemmas 11–13 and 14 given here). A machine-checked (Lean/Mathlib) formalization of this argument has also been completed, as a separate, complementary effort, available at [7]; brute-force simulation and independent reading catch different classes of error than a machine-checked proof does, and vice versa.

5. Related work

The reformulation via φ_d and the resulting rigidity phenomenon (Lemma 8) form the natural two-dimensional analogue of the classical theory of *covering systems of congruences*, introduced by Erdős [3] to show that a positive proportion of the integers are not of the form $2^k + p$ for p prime. A very close relative of our covering problem — covering $\mathbb{Z}^2$ by finitely many finite-index *sublattices* (necessarily containing the origin, unlike the general affine cosets considered here) — is studied by Cremona and Koymans [1], who give it the same "projective/homogeneous covering congruences" framing; their refinement machinery (splitting one lattice into several of prime-power relative index) is structurally reminiscent of the splitting used here, one rank up.

A different classical result addresses a related but distinct question about how many directions a covering needs. Corzatt [2] conjectured that if a *finite, convex* set of lattice points is covered by n lines (with no constraint on where the lines may cross), the lines can always be chosen to use at most four distinct slopes. Our setting is the opposite regime: an infinite, unbounded point set, with a crossing-avoidance constraint standing in for a bound on the number of lines, and correspondingly the answer is qualitatively different: densely many directions are both necessary to consider and available.

The unimodular pairs forbidden by Lemma 8 are exactly the Farey-neighbor pairs of slopes; see e.g. Hardy and Wright [4, Ch. III] for the classical theory of Farey sequences and the closely related Stern–Brocot tree. This is precisely the toolkit used in the digital-geometry literature on digitized straight lines, where continued-fraction and Farey-tree techniques describe digital approximations to a line of given (rational or irrational) slope; see Kiselman [5], and, for the continued-fraction approach specifically, Uscka-Wehlou's dissertation [6].

Open problem 17. Is it possible to generalize Theorem 7 to $\mathbb{Z}^n$ for $n \geq 3$? In this formulation, one would study configurations of "lattice hyperplanes" or "lattice lines", such that no two different directions meet at a lattice point, and determine which sets of directions can occur.

Acknowledgements

This problem occurred to the author during an idle conversation with colleagues (in the department coffee room) about covering the lattice $\mathbb{Z}^2$ with lines. It was considered as a useful test for LLM-assisted proof discovery and proof formalization, since the author conjectured that the proof of the result, if it was indeed true, should only use elementary methods, which are already in lean (Mathlib).

The results in this note: the reformulation, the rigidity and coset lemmas, the recursive construction, and both rounds of error-correction (one caught by numerical simulation, one by an independent review), were found by Claude (Anthropic), working with the author in an interactive session. The additional references collected in the extended-references version of this note were located and verified by Claude via web search. A list of "certificates" showing how each reference is used in this manuscript (this "certificate of non-hallucination"), together with a more detailed explanation of the whole process of producing this manuscript, from inception to exploration to proof discovery to proof formalization and verification, are described in the gitlab repository [7].

This manuscript was likewise produced by the author and Claude jointly, with the author mainly directing Claude toward clearer exposition for human readers: restructuring the proofs into short, explicitly labeled lemmas, and breaking up long equations for readability. The author then checked the final manuscript to the best of his ability, made final edits, and assumes full responsibility for any errors, mistakes, or gaps that remain.

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Conflicts of interest

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